

Practice Set: Number System MCQs

Mathematics Series

December 30, 2025

1 Instructions

This document contains 30 multiple-choice questions designed for competitive exam preparation. Choose the most appropriate option for each question. Solutions are provided at the end of the document.

2 Questions

1. What is the smallest prime number?

- A) 0 B) 1 C) 2 D) 3

2. The sum of the first 50 natural numbers is:

- A) 1225 B) 1275 C) 1300 D) 2550

3. Which of the following numbers is divisible by 9?

- A) 12345 B) 67812 C) 45306 D) 54321

4. What is the unit digit in the product $(7^{71} \times 6^{63} \times 3^{65})$?

- A) 1 B) 2 C) 4 D) 6

5. If a number is divisible by both 11 and 13, then it must be necessarily divisible by:

- A) $(11 + 13)$ B) $(13 - 11)$ C) (11×13) D) None of these

6. How many prime numbers are there between 1 and 100?

- A) 23 B) 24 C) 25 D) 26

7. What is the remainder when 2^{31} is divided by 5?

- A) 1 B) 2 C) 3 D) 4

8. A number when divided by 899 gives a remainder 63. If the same number is divided by 29, the remainder will be:

- A) 2 B) 3 C) 5 D) 10

9. The difference between the place value and the face value of 6 in the numeral 856421 is:

- A) 0 B) 5994 C) 6000 D) 5400
- 10.** Which of the following is a rational number?
- A) $\sqrt{2}$ B) π C) 0.121212... D) $\sqrt{3}$
- 11.** The sum of two numbers is 40 and their difference is 4. The ratio of the numbers is:
- A) 11:9 B) 10:7 C) 22:18 D) 11:8
- 12.** Find the largest number of four digits exactly divisible by 12, 15, 18, and 27.
- A) 9600 B) 9720 C) 9930 D) 9960
- 13.** What is the value of $(0.333... + 0.444...)$?
- A) 0.7 B) $7/9$ C) $7/10$ D) $7/99$
- 14.** If n is an even number, then $(6^n - 1)$ is divisible by:
- A) 35 B) 30 C) 6 D) 37
- 15.** The total number of even prime numbers is:
- A) 0 B) 1 C) 2 D) Infinite
- 16.** How many digits are required to number a book containing 200 pages?
- A) 492 B) 392 C) 500 D) 480
- 17.** Which is the largest among the following fractions: $2/3$, $3/5$, $8/11$, $11/17$?
- A) $2/3$ B) $3/5$ C) $8/11$ D) $11/17$
- 18.** The product of two consecutive even numbers is 528. What are the numbers?
- A) 20, 22 B) 22, 24 C) 18, 20 D) 24, 26
- 19.** If $1^2 + 2^2 + 3^2 + ... + 10^2 = 385$, then the value of $2^2 + 4^2 + 6^2 + ... + 20^2$ is:
- A) 770 B) 1155 C) 1540 D) 1500
- 20.** The sum of three consecutive odd numbers is 57. The middle number is:
- A) 17 B) 19 C) 21 D) 23
- 21.** A number is as much greater than 36 as is less than 86. The number is:
- A) 50 B) 61 C) 62 D) 65
- 22.** What is the smallest number by which 675 must be multiplied to make it a perfect cube?
- A) 2 B) 3 C) 5 D) 6
- 23.** The L.C.M. of two numbers is 48. The numbers are in the ratio 2:3. The sum of the numbers is:

- A) 28 B) 32 C) 40 D) 64
- 24.** Which of the following is not a prime number?
- A) 31 B) 61 C) 71 D) 91
- 25.** The sum of the digits of a two-digit number is 9. If 27 is added to the number, its digits are reversed. The number is:
- A) 27 B) 36 C) 45 D) 54
- 26.** What is the value of $(1 - 1/2)(1 - 1/3)(1 - 1/4) \dots (1 - 1/n)$?
- A) $1/n$ B) $2/n$ C) $(n - 1)/n$ D) $1/(n - 1)$
- 27.** The sum of all prime numbers between 30 and 50 is:
- A) 150 B) 161 C) 199 D) 200
- 28.** If x and y are odd numbers, then which of the following is even?
- A) $x + y$ B) $x + y + 1$ C) xy D) $xy + 2$
- 29.** The number of factors of 100 is:
- A) 6 B) 7 C) 8 D) 9
- 30.** What is the value of $\sqrt{12 + \sqrt{12 + \sqrt{12 + \dots}}}$?
- A) 3 B) 4 C) 6 D) 12

3 Answer Key and Explanations

1. **C** — 2 is the only even prime and the smallest.
2. **B** — Formula: $n(n + 1)/2 \rightarrow 50 \times 51/2 = 1275$.
3. **C** — Sum of digits $4 + 5 + 3 + 0 + 6 = 18$, which is divisible by 9.
4. **C** — Unit digit cycle analysis results in 4.
5. **C** — Since 11 and 13 are co-prime, the number must be divisible by their product.
6. **C** — There are exactly 25 prime numbers between 1 and 100.
7. **C** — $2^1 = 2, 2^2 = 4, 2^3 = 8 \equiv 3, 2^4 = 16 \equiv 1 \pmod{5}$. Remainder for 2^{31} is 3.
8. **C** — $63 \pmod{29} = 5$.
9. **B** — $6000 - 6 = 5994$.
10. **C** — Repeating decimals like 0.1212... can be expressed as $12/99$.
11. **A** — $x + y = 40, x - y = 4 \implies x = 22, y = 18$. Ratio $22 : 18 = 11 : 9$.
12. **B** — $LCM(12, 15, 18, 27) = 540$. Largest 4-digit multiple is 9720.
13. **B** — $3/9 + 4/9 = 7/9$.
14. **A** — If $n = 2, 6^2 - 1 = 35$.
15. **B** — Only '2' is an even prime number.
16. **A** — $(9 \times 1) + (90 \times 2) + (101 \times 3) = 9 + 180 + 303 = 492$.
17. **C** — $8/11 \approx 0.72$, which is higher than $2/3(0.66)$ and others.
18. **B** — $22 \times 24 = 528$.
19. **C** — $4 \times (1^2 + \dots + 10^2) = 4 \times 385 = 1540$.
20. **B** — $x + (x + 2) + (x + 4) = 57 \implies 3x = 51 \implies x = 17$. Middle is $17 + 2 = 19$.
21. **B** — Average: $(36 + 86)/2 = 61$.
22. **C** — $675 = 3^3 \times 5^2$. Needs one more 5 to become 5^3 .
23. **C** — Numbers are 16 and 24. Sum = 40.
24. **D** — 91 is 13×7 .
25. **B** — $36 + 27 = 63$.
26. **A** — Telescoping product results in $1/n$.
27. **C** — $31 + 37 + 41 + 43 + 47 = 199$.
28. **A** — Sum of two odds is always even.
29. **D** — $(2 + 1) \times (2 + 1) = 9$ factors.
30. **B** — $x^2 - x - 12 = 0 \implies (x - 4)(x + 3) = 0$. $x = 4$.